Activity 1: Exploring Computational Thinking Through Bebras Cards

Rationale:

Why are PSTs expected to learn this? Why are the outcomes of this lesson important? Why are these outcomes essential for future learning?

PSTs need an initial understanding of computational thinking to recognise and purposefully address CT in their future mathematics teaching. Many CT processes are closely connected to ways of thinking already present in mathematics, and making these connections explicit can help PSTs understand how mathematical tasks can provide opportunities for developing CT. Learning to identify and describe CT processes also gives PSTs a language for analysing tasks and recognising which CT processes pupils engage in when solving them. This understanding provides a foundation for later selecting, adapting, and designing mathematical tasks and teaching activities that intentionally support pupils' CT development.

Aims and objective:

Towards which goal are PSTs expected to learn?

PSTs will develop an initial/intuitive understanding of CT and selected CT processes, recognising how they are involved in task-solving, and exploring their connections to mathematical thinking and mathematics teaching.

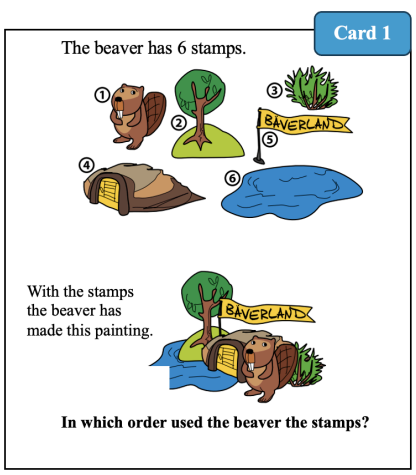
Time:

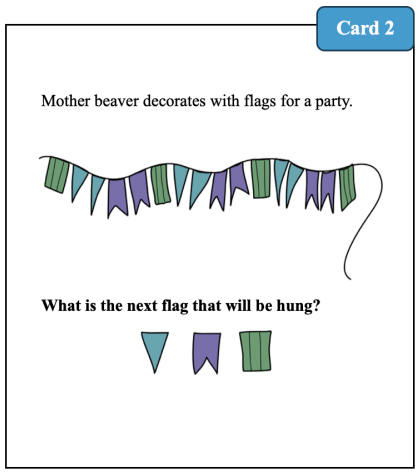
How long is the lesson?
How do you divide the time?

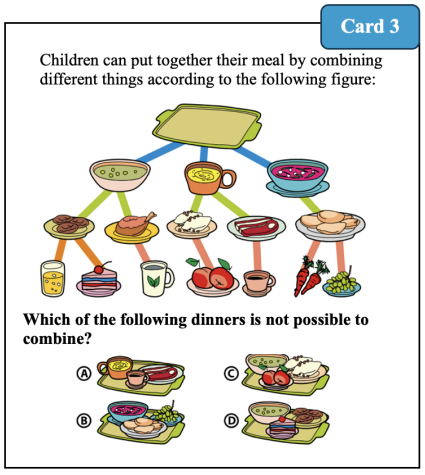
In total, approximately 120 minutes, depending on PSTs previous experiences and knowledge.

Step 1: 15-20 minutes
Step 2: 20-30 minutes
Step 3: 20-30 minutes
Step 4: 20-30 minutes
Step 5: 10 minutes

Content/Topic: Computational thinking through unplugged tasks (cards)	**Assessment:** How will you know that you reached the lesson's objectives? How will PST learning, in the short terms, be assessed? The assessment in this lesson comes in two levels: ● Formative assessment during the whole-class discussion in STEP 5. TE can have an overall idea of how well PSTs understood the mentioned CT components and CT itself. ● Self-assessment for PSTs in the form of exit cards. Exit cards are provided at the end of this document.
Learning activities: How are PSTs expected to learn this? What specific activities, investigations, problems, questions, or tasks will pupils be working on during the lesson? PSTs are going to work in groups of 2-4 in five steps.	**PSTs prior knowledge:** What prior knowledge do PSTs need? PSTs need to be familiar/know the expected learning outcomes in mathematics for the grades PST will be qualified to teach.







STEP 1 - HIDDEN PROCESSES

In this step, PSTs are going to work with three (Bebras) cards. PSTs first solve each task displayed on the cards individually and then compare their solutions and strategies within their group. The cards are addressed one at a time. After each task (Bebras Cards 1, 2, and 3), the TE facilitates a whole-class discussion using the same two questions:

- How do you know your answer is correct?
- What is hidden in the task? What do you think the task conveys?/What ideas or ways of thinking do you think the task is intended to convey?

The purpose of these questions is to encourage PSTs to become aware of their own thinking while solving the tasks, including the strategies and processes they use. The discussion should initially remain open and draw on PSTs' own descriptions of these processes, without introducing the CT processes beforehand. At the end of the discussion of each card, the TE makes the relevant CT process(es) explicit, introduces and briefly explains the corresponding terminology, and connects it to the processes identified by the PSTs. In this way, PSTs gradually develop a shared language for describing the thinking processes that are associated with CT.

CARD 1: The discussion is expected to highlight *sequencing/algorithmic thinking*.

- *Sequencing/algorithmic thinking:* PSTs need to determine the order in which the stamps were used. They can use the overlaps between elements to infer which actions must have occurred before or after others and reconstruct the sequence of actions. The order matters: if the stamps are applied in a different order, the resulting picture will also be different.

CARD 2: The discussion is expected to highlight *pattern recognition*.

- *Pattern recognition:* PSTs need to identify the repeating pattern in the sequence of flags and use this pattern to determine which flag comes next. In general, pattern recognition involves looking for regularities in the given information and using them to make a prediction.

CARD 3: The discussion is expected to highlight *evaluation* and *debugging*.

- *Evaluation:* PSTs need to check the proposed dinners against the rules represented in the diagram and determine which combinations satisfy these conditions and which do not.
- *Debugging:* After identifying an invalid dinner, PSTs can locate the source of the problem by determining which part of the combination violates the rules represented in the diagram. Note: this is not expected in the task, as the question "only" is: Which of the dinners is not possible to combine. However, the debugging might naturally occur as part of justification of why the identified dinner can't be combined.

STEP 2 - OTHER POTENTIAL CT PROCESSES

In this step, PSTs redesign one of the cards used in Step 1 so that the redesigned task involves one or more of the other CT processes discussed in Step 1, in addition to those already involved in the original task. Each group selects one card and has approximately 5-7 minutes to modify the task. The aim is not simply to make the task more difficult, but to change it in a way that creates opportunities for other CT processes to be involved in the solution process.

After completing their redesign, groups present their redesigned cards to another group. As part of the presentation, they explain which CT processes are involved in their redesigned task and justify how they are reflected in the task and its solution process.

The TE can organise the cross-group presentations in two ways:

- Groups that redesigned the same card can be paired, allowing PSTs to compare different ways of extending the same original task.
- Groups that redesigned different cards can be paired, allowing PSTs to encounter a wider range of tasks and CT processes.

After the group presentations, a brief whole-class wrap-up led by the TE can follow, perhaps highlighting one or two interesting redesigns or discussing whether the claimed CT processes are actually involved.

STEP 3 - APPLICATION FOR TEACHING CT AND MATHEMATICS

In this step, PSTs will apply what they have learned in Steps 1 and 2. Their task is to design a mathematics task for a grade level and mathematical topic of their choice. The task should provide opportunities for pupils to engage with one or more of the CT processes discussed in the previous steps.
PSTs should consider:

- Which mathematical topic and grade level is the task intended for?
- Which CT process(es) should be involved in solving the task? Why is the chosen mathematical topic suitable for this integration?
- What will pupils need to do in order to solve the task, and how does this engage the intended CT process(es)?

TE selects 3–4 tasks, ideally representing different mathematical topics and/or CT processes. For each, the group briefly explains their answers to the above-listed questions.

TE facilitation in case of PSTs find it difficult to connect CT processes to mathematics:
Sequencing/algorithmic thinking (card 1): In mathematics, one can find similar ideas in geometry, where a construction consists of a sequence of steps that need to be carried out in an appropriate order. The order can also matter when applying geometric transformations, as performing transformations in a different sequence may result in a different final image. Similar ideas occur in arithmetic and algebra: written calculation algorithms involve carrying out operations in a particular sequence, and solving an equation involves a sequence of operations in which each step builds on the previous one.

Pattern recognition (card 2): In mathematics, one can find similar ideas when working with number and shape patterns and sequences, where identifying a regularity allows you to predict subsequent elements or formulate a rule describing the pattern. Pattern recognition is also important in algebra, where observed regularities can provide a basis for generalisation, for example, moving from several particular cases to expressing a relationship using variables.

Evaluation, debugging (card 3): In mathematics, one can find similar ideas when checking solutions and identifying errors. For example, a proposed solution to an equation can be evaluated by substituting it into the original equation and checking whether it satisfies the equation. When analysing an incorrect solution, individual steps can be examined to identify where an error occurred and why. **STEP 4 – From mathematical thinking to CT** In this step, PSTs move from considering individual CT components to developing an initial understanding of computational thinking and its relationship to mathematical thinking. This will be done in three subtasks. a) The TE presents the following task: *Calculate 37 x 24. How did you calculate it? Be prepared to explain the strategy you used.* PSTs first solve the task individually and then briefly compare their strategies. The TE may invite several PSTs to describe how they obtained their answer. Different strategies are possible and should be welcomed [e.g., standard written algorithm, (30 + 7) x (20 + 4), doubling/halving]. The purpose at this point is simply to make the solution processes visible. b) The TE then changes the task: *Develop and describe a method that works for multiplying any two 2-digit whole numbers.* PSTs discuss in their group how their particular strategy would need to be adapted or expressed as a more general procedure. For example, if a group used a strategy based on decomposition and the distributive property (37 x 24) = (30 + 7) x (20 + 4) = 30 x 20 + 7 x 20 + 30 x 4 + 7 x 4 = 600 + 140 + 120 + 28 = 888, then the corresponding general algebraic procedure could be expressed as: (10a + b) x (10c + d) = 10a x 10c + b x 10c + 10a x d + b x d. If a group used an area model of multiplication, the corresponding general procedure could be visualized as:	



Again, the TE invites groups with different approaches to present their general procedures on the blackboard.

c) Finally, the TE introduces an additional requirement:
"Imagine that the multiplication will be carried out by someone else, and not you. Write instructions that another person (who doesn't know how to multiply 2-digit numbers) or a computer machine could follow to multiply any two positive whole 2-digit numbers. What would your instructions need to be like?"

PSTs work on the subtask in groups for a few minutes. The purpose is not for PSTs to develop formally complete multiplication instructions/algorithm. Rather, the Step 4 activity should make visible the shift from knowing how to solve a particular mathematical problem to expressing a solution method in a form that can be transferred to and carried out by another agent.

After PSTs have worked on the instructions, the TE can ask:

- What changed when you had to formulate the method for someone/something else to carry it out rather than for yourselves?
- What characteristics do/would your instructions need to have? *(e.g., precise, unambiguous, systematic, ordered)*
- Which of the thinking processes/components discussed earlier in the lesson (in steps 1-3) can you recognise here? (e.g., *sequencing/algorithmic thinking, evaluation*; the TE can also draw attention to *abstraction* and *generalization*)

The TE uses PSTs' responses to introduce computational thinking as the concept that describes students' solutions to the subtasks. CT can be presented as a way of approaching and solving problems systematically, drawing on

processes such as those encountered in the lesson, with particular attention to structuring and expressing problems and their solutions in ways that allow the solution process to be carried out by another human or by a computer machine.

The TE emphasises that the processes encountered during the lesson are not unique to CT - pattern recognition, algorithmic thinking, abstraction, generalisation and evaluation, for example, are also familiar from mathematical thinking. This substantial overlap between mathematical and computational thinking is one reason why mathematics provides a meaningful context for developing CT. What brings a specifically computational perspective is the attention to how a problem and its solution can be structured and expressed so that the solution process can be systematically transferred and carried out.

STEP 5 – CLOSURE

To close the lesson, the TE returns briefly to the three cards used in Step 1 and the mathematical tasks designed by PSTs in Step 3 and asks:

> *Looking back at today's activities, how would you now explain computational thinking to another mathematics teacher? What is the relationship between CT and mathematical thinking?*

The discussion should lead to the recognition that teaching CT through mathematics does not necessarily mean introducing entirely new ways of thinking. Mathematics already provides opportunities for many of the processes associated with CT. Developing CT can involve making these processes visible and deliberately creating opportunities for pupils to use them, while extending their problem solving towards more explicit, systematic and potentially executable ways of formulating solutions.

Finally, the TE asks PSTs to complete the exit cards as part of their self-assessment. The cards will also provide the TE with feedback on PSTs' learning.

Grouping and Location:

With whom are PSTs expected to learn? Is grouping going to vary during the lesson? Where are PSTs learning?

PSTs are expected to work both individually and in groups, thinking didactically by reflecting on their own learning experiences in order to better understand how their pupils may approach and experience problems involving CT.

Teacher educator's facilitation:

How is the teacher educator facilitating PSTs' learning?

The TE elicits rather than initially names CT processes, using probing questions such as "How did you think?" and "What makes this process necessary for solving the task?". The TE selects and contrasts different PST strategies, asks PSTs to justify their identification of CT processes, and introduces CT terminology after the processes have emerged from discussion. When needed, the TE provides mathematical examples or prompts. Finally, the TE connects PSTs' experiences across the activities and makes explicit the relationship between mathematical and computational thinking.

Material and Resources:

What materials and/or resources are essential/beneficial for PSTs (and the teacher educator) to successfully accomplish the learning goals? And why?

- For this lesson, we use 3 translated Bebras cards (no. 4, 7, 17) from *BEBRAS. Upptäcka Datalogist tänkande*.
- If possible, use paper versions of Bebras cards. Some countries have them printed out; some of the examples are online as well. If you don't have access to the original Bebras cards, you can just print the attached files or add them to your slides.
    - You can read more about Bebras here: https://www.bebras.org/ .
    - English version of Bebras cards: https://www.bebras.uk/index.php?action=content&id=37
    - Different Bebras products (check the language before ordering): https://bebrasplay.com/en/
    - Different Bebras printables for free: https://bebrasplay.com/en/content/11-for-free
    - Bebras tasks for 365: https://amt.edu.au/bebras-365
    - Bebras tasks in Estonian: https://kobras.eio.ee/

Follow up activities

What questions/activities are appropriate for the PSTs to do after the lesson?/ What activity will reinforce the learning goals?

A number of activities can be used to expand and learn more about the CT processes

- More detailed description of CT, its numerous definitions, and CT processes could be added.
- In this lesson, only 3 tasks from Bebras are chosen. To include more CT processes and make connections between them, other Bebras cards could be used. See the links above to find more cards for similar activities.
- After Step 3, one could include a more thorough analysis/discussion of the new tasks PSTs proposed. They could be checked from different viewpoints (national curricula: is the task suitable for the chosen class; understanding of CT processes and their suitability for the mathematical task; etc.)
- The PSTs can compare their new tasks (from Step 3) and discuss benefits and drawbacks with each task.
- In Step 1, an additional video describing the similar idea as in Card 1 could be used. You can find the video here: https://youtu.be/FN2RM-CHkuI?si=ANcG8q7wP1-U__AA The crucial part is to make clear that how you give directions, how clear the instructions are, in what order are they given, plays a crucial role.

Exit Card

Date:

Student name:

---

1. Select the smiley that best describes your overall feeling about your learning today.



Explain why you selected this smiley:

2. Describe the most important thing(s) you learned today.

3. Which part of the session did you like best today? Explain why. /Describe the activity you liked the best today and explain why.

4. Describe what you found difficult today, and explain why.

5. Select the smiley that best describes how confident you feel about

- understanding the core ideas of computational thinking 
- understanding the core ideas of CT processes 

Explain why you selected these smileys: